Introduction

The prevalence of hydrometeorological events poses a serious health risk to many countries in Latin America. In the past couple of decades, over 200 disasters have been recorded, affecting approximately 18 million inhabitants (Alcántara-Ayala, 2019). These natural hazards, such as floods and droughts, are typically associated with El Niño – a seasonal climate pattern whose effects can differ in magnitude across regions of the country. The groups most vulnerable to these risks are the impoverished and the young (Philipsborn and Chan, 2018). Moreover, these risks impose health shocks that can have both short-run and long-run effects. In children, these shocks can lead to stunting and wasting. These outcomes matter later in life, as taller individuals have been found to earn higher wages (Vogl, 2014). They can also operate through other channels such as educational attainment and cognitive and socioemotional development.

Research on how events during early developmental stages (the fetal period, infancy, and childhood) shape long term health is well established. This idea is known as the "Fetal Origins of Adult Disease" or "Barker's hypothesis" (Barker 1998). Many scholars have contributed evidence supporting this hypothesis, finding impacts on educational attainment, physical disability, income, and socioeconomic status (Almond 2006; Maccini and Yang 2009).

Less research in this space has examined medium-term effects, or the period between birth and adulthood (Almond et al., 2018). Along with several earlier studies, my research seeks to contribute to this area (Webb 2024; Thamarapani 2021; Aguilar and Vicarelli 2022). First, my research contributes by using a constructed dataset that combines spatial weather data with the Mexican Family Life Survey (MxFLS). This allows me to track an individual's health status across three survey waves, from birth to adolescence. Second, most prior literature does not address how household-level insurance mechanisms may counteract the impact of natural disasters on child health outcomes. As access to credit and remittances helps relieve liquidity constraints (Deaton, 1991), I analyze whether these resources moderate the effect of shocks on child health outcomes. Finally, I aim to add onto prior research by using several specifications of remittances and natural disasters. I distinguish between domestic and international remittances. Because the agroclimatology literature finds that floods have little effect on crop production (Lesk et al., 2016), I focus mainly on droughts as my natural disaster shock. The drought specifications include a single major drought, multiple droughts, and a continuous rainfall measure.

This leads to our research questions. First, does exposure to drought during the prenatal period or early childhood affect children's anthropometric outcomes such as height-for-age (HAZ) and weight-for-age (WAZ) z-scores? Second, does access to credit or remittances mitigate the adverse effects of drought on children's health?

Contribution

The literature on this topic is extensive. Several studies in Mexico use the Mexican Family Life Survey to examine how shocks such as crop loss, the Mexican drug war, exposure to air pollution, and migration affect childhood health outcomes (Michaelsen and Tolan 2012; Brown 2018; Molina 2020; Damon and Kristiansen 2014; Farfan et al. 2012). However, none

of these studies specifically analyze the impact of natural disasters on childhood outcomes. They also do not use remittances, likely because the survey has no explicit remittance variable and it must be constructed. More closely related research in Mexico relies on cross-sectional data or non-representative samples of poor, marginalized communities. In contrast, I use a nationally representative longitudinal panel that allows me to track the same child over time (Aguero 2014; Skoufias and Vinha 2012; Adhvaryu et al. 2018; Aguilar and Vicarelli 2022). Additionally, some papers in Mexico test whether government programs such as Progresa, Liconsa, and Procampo help mitigate climate shocks, whereas I focus on private and household-level buffers such as credit and remittances. Many studies outside of Mexico also exist, but many of them focus on a single large shock. These papers' findings may not generalize to smaller and more frequent shocks (Caruso and Miller 2015; Fujiki and Hsiao 2015; Deuchert and Felfe 2015; Dodlova et al. 2023). My analysis incorporates one-time, multiple, and continuous shocks and tests whether the results hold across all specifications.

Perhaps most similar to my study are those that use the Indonesian Family Life Survey (IFLS), a longitudinal panel survey conducted by RAND. Similarly, these studies analyze the impact of rainfall on health, schooling, and cognitive and socioemotional development (Webb 2024; Maccini and Yang 2009; Adams and Cuecuecha 2010; Thamarapani 2021). Thamarapani (2021) also investigates whether credit and remittances can buffer childhood health outcomes. However, she relies on self-reported categorical disaster data that is subject to missing magnitudes, mismeasurement, recall error, and self-reporting bias. I improve on this by merging in gridded rainfall time-series data. Webb (2024) focuses on adult cognition and social behavior, and Adams and Cuecuecha (2010) focus on household consumption and poverty, whereas I emphasize childhood anthropometric outcomes. Unlike Maccini and Yang (2009), I am looking at a medium time horizon rather than the long-run effects of a shock from birth to adult health outcomes. In addition to the different contexts of all these studies, I will utilize a different empirical strategy, with DID as the main specification. My results can therefore test whether their findings hold in a different setting and under a different methodology, while also shedding light on the role of remittances as informal insurance.

Data Sources

The Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) and the Mexican Family Life Survey (MxFLS) are used for this analysis. The MxFLS is converted into a longitudinal panel dataset and is representative of the Mexican population at the national level, covering both urban and rural individuals. It spans a 10-year period across three rounds: 2002, 2005–2006, and 2009–2012. CHIRPS provides rainfall variation in millimeters across Mexico. It offers over 35 years of data, spanning from 1981 to the present, and generates a gridded rainfall time series.

I constructed a dataset that combines CHIRPS spatial weather data with the MxFLS data. Simply put, I link these datasets by overlaying CHIRPS gridded precipitation data on INEGI municipal boundaries, computing area-weighted annual rainfall for each municipality, and merging it with MxFLS households by municipality and interview year.

Sample Composition

This section is unfinished.

Additional Notes

- The proxy for domestic/international remittances is constructed from MxFLS Book IIIB and captures money received from parents, siblings, and children living outside the household. Coded 1 as receiving remittances if any household member aged 15 or older received cash for health, school tuition, or other purposes. Responses from the proxy book are included to capture household members that were not interviewed.
- The proxy for international remittances is U.S. specific. A household is coded as receiving U.S. remittances if any member received help in cash/other from a mother and/or father whose reported country of residence is the United States. Captures only parental transfers due to data limitations.
- Municipalities are formed by combining the household's state and municipality of residence at the time of interview. These are combined into the five-digit INEGI key (two-digit state plus three-digit municipality). A small number of codes that do not appear in the INEGI catalog are corrected using the same household's municipality in other waves or set to missing.

Variables utilized

These are not set in stone.
These controls will be utilized or considered:

- **Outcome:** HAZ, WAZ, BMIZ, hemoglobin, education, and cognitive and socioemotional development measures
- **Exposure:**
  - The 2008 shock, which falls between the two waves, captures municipalities that experienced a drought in 2008. Drought is defined relative to each municipality's own rainfall history, as rainfall below the 20th percentile of that municipality's distribution (Shah and Steinberg 2013).
  - Number of natural disaster types, Number of droughts
  - Continuous log rainfall
- **Pre-disaster controls:**
  - **Child Characteristics:** Gender, birth year, birth month, birth order, oldest child, birth weight, delivered at home, weeks breastfed, anemic

- **Parent Characteristics:** Prenatal visits to health facility (need to check if I have), mother's height and weight, father's height and weight, mother's age at childbirth, father's age at child birth, mother employed, father employed, mother years of education, father years of education
- **Household characteristics:** locality, rural vs urban, rooms per person, availability of good floor quality, material of roofing, good ventilation, electricity, television, phone

Empirical Methodology

These are not set in stone.

This model estimates the effect of drought on child health outcomes using a difference-in-differences design. Below, we include the specification with household buffers as controls:

$$Y_{imt} = \beta \left(\text{Drought}_m^{2008} \times \text{Post}_t\right) + \sum_k \gamma_k \left(B_{h,k}^{2002} \times \text{Post}_t\right) + \text{Post}_t \times (\mathbf{C}_i'\boldsymbol{\theta} + \mathbf{P}_h^{2002\prime}\boldsymbol{\phi} + \mathbf{X}_h^{2002\prime}\boldsymbol{\delta}) + \alpha_i + \lambda_t + \varepsilon_{imt} \tag{1}$$

where Y_{imt} is the height-for-age (HAZ) or weight-for-age (WAZ) z-score of child i, living in municipality m at baseline, observed in survey wave t. Anthropometrics are measured before the shock at baseline in MxFLS wave 2 (2005–2007) and are measured after the shock in MxFLS-3 (2009–2013). Drought_m^{2008} equals one if annual rainfall in 2008 in the child's 2005 municipality fell below the 20th percentile of that municipality's own 1981–2004 rainfall distribution (Shah and Steinberg, 2013). Post_t equals one for observations interviewed after 2008, i.e. in MxFLS-3. The 2008 drought falls strictly between the two survey waves, MxFLS-2 and MxFLS-3 so no child is interviewed during the shock year. The coefficient of interest, β, captures the average change in child anthropometrics in drought-affected municipalities relative to unaffected municipalities.

$B_{h,k}^{2002}$ denotes the two household buffers. That is (i) whether the household received remittances and (ii) whether a formal bank was present in the community. $\mathbf{X}_h^{2002}$ is a vector of household controls, $\mathbf{C}_i$ is a vector of child characteristics and $\mathbf{P}_h^{2002}$ is a vector of adult characteristics. Both the buffers and the controls are measured in MxFLS-1 (2002), before the drought. α_i are child fixed effects. λ_t are survey-wave fixed effects. HAZ is estimated for children under 16 and WAZ for children under 11, following the WHO reference ranges.

There are a number of alternative specifications I will look into. These include a triple-difference design that adds remittances, an IV for remittances, different ways of defining the remittance and rainfall variables, and rainfall measured before birth, at birth, and after birth. etc.

Current Progress

The linkage between CHIRPS and MxFLS is complete at the municipality-year level. I have constructed remittance proxies for international remittances alone and for domestic

and international remittances combined. Some controls have already been added, and the remaining necessary ones are in progress. The Ponnusamy-style regressions, DiD, and DDD baseline regressions are done and are now being finalized before I send them out.

Citations

- https://www.investigacionesgeograficas.unam.mx/index.php/rig/article/view/60025
- https://publications.aap.org/pediatrics/article-abstract/141/6/e20173774/37657/Climate-Change-and-Global-Child-Health?redirectedFrom=fulltext
- https://www.sciencedirect.com/science/article/abs/pii/S0304387813001697
- https://pubmed.ncbi.nlm.nih.gov/9680492/
- https://www.amazon.es/Mothers-Babies-Health-Later-Life/dp/0443061653
- https://www.nature.com/articles/nature02725
- https://www.jstor.org/stable/10.1086/507154?seq=1
- https://www.aeaweb.org/articles?id=10.1257/aer.99.3.1006
- https://www.aeaweb.org/articles?id=10.1257/jel.20171164
- https://academic.oup.com/ej/article/134/660/1637/7455874
- https://pubmed.ncbi.nlm.nih.gov/26738594/
- https://www.nber.org/papers/w3196
- https://www.sciencedirect.com/science/article/pii/S0305750X21003053
- https://www.nber.org/papers/w24848
- https://ageconsearch.umn.edu/record/165044
- https://www.sciencedirect.com/science/article/pii/S1570677X11000724
- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2199465
- https://pubmed.ncbi.nlm.nih.gov/29344926/
- https://onlinelibrary.wiley.com/doi/10.1111/agec.12117
- https://academic.oup.com/jeea/article/19/1/580/5802826
- https://www.cedlas.econo.unlp.edu.ar/wp/wp-content/uploads/farfan-paper.pdf
- https://www.sciencedirect.com/science/article/abs/pii/S0304387815000917
- https://www.sciencedirect.com/science/article/abs/pii/S0304407614002541
- https://www.sciencedirect.com/science/article/abs/pii/S0014292115001427

- https://docs.iza.org/dp16195.pdf
- https://www.rand.org/health/surveys/FLS/IFLS.html
- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1683904
- https://www.cambridge.org/core/journals/environment-and-development-economics/article/natural-disasters-and-child-health/4D0ED2F35C4721541DD146BF490D3F50
- https://www.chc.ucsb.edu/data/chirps
- https://www.ennvih-mxfls.org/english/index.html